Python DSA — Arrays & Algorithms

Complete Practice Sheet | Foundation + 6 Algorithms + Mixed Problems | 70 Questions

How to Use This Sheet

- Pehle theory padho (2-3 minutes) — concept samjho.
- Phir Easy questions solve karo — foundation solid karo.
- Medium aur Hard tab karo jab Easy confident feel ho.
- Stuck ho toh Hint dekho — 15 min baad — phir try karo.
- Har algorithm ke baad Mixed Problems mein jaao.

PART 1 — Array Foundation

Key Concepts

- Array (Python mein List) = ek ordered collection jisme ek se zyada values store hoti hain.
- Indexing: arr[0] = pehla element, arr[-1] = aakhri element. O(1) time.
- Slicing: arr[a:b] = a se b-1 tak ke elements. arr[::-1] = reverse.
- Common operations: append() O(1), insert() O(n), pop() O(1), remove() O(n).
- Length: len(arr). Traverse: for i in arr ya for i in range(len(arr)).
- 2D Array: list of lists — arr[row][col] se access karo.

Practice Questions

#	Question	Hint	Level
1	Ek list create karo 1-10 numbers ki aur print karo.	*list(range(...))*	Easy
2	List ka pehla aur aakhri element print karo.	*arr[0], arr[-1]*	Easy
3	List mein ek element insert karo specific index pe.	*arr.insert(i, val)*	Easy
4	List se ek specific value delete karo.	*arr.remove(val)*	Easy
5	List ko reverse karo bina reverse() use kiye.	*slicing ya loop*	Easy
6	List ke elements ko double karo (har element * 2).	*loop ya list comp*	Easy
7	Do lists ko merge karo ek naye list mein.	*loop se append*	Easy
8	List mein kitne positive aur kitne negative numbers hain count karo.	*2 counters*	Easy
9	List ke saare even-index elements print karo.	*range(0,n,2)*	Easy
10	List mein second largest element dhundo bina sort kiye.	*max track karo*	Medium
11	List ke elements ka running sum banao (prefix sum array).	*prefix[i]=prefix[i-1]+arr[i]*	Medium
12	List mein saare duplicates dhundo aur print karo.	*seen set*	Medium

13	List ko even numbers pehle, odd baad mein arrange karo (order maintain karo).	*2 separate lists*	Medium
14	3x3 matrix banao aur uski transpose nikalo.	*arr[j][i]*	Medium
15	List mein longest consecutive sequence ki length nikalo. (e.g. [1,2,3,7,8] → 3)	*sort + count*	Hard

□ ALGORITHM 1 — Linear Search

Key Concepts

- Linear Search = har element ek ek check karo jab tak target na mile.
- Time Complexity: O(n) — worst case poori list traverse karni padti hai.
- Space Complexity: O(1) — koi extra space nahi chahiye.
- Kab use karein: Unsorted array mein, ya chota data hone pe.
- Return karo: index agar mila, -1 agar nahi mila.

Practice Questions

#	Question	Hint	Level
16	List mein ek given number ka pehla index dhundo, -1 return karo agar na mile.	*loop + compare*	Easy
17	List mein ek element kitni baar aata hai count karo (Linear Search se).	*counter++*	Easy
18	List mein saare occurrences ke indexes return karo.	*result list mein append*	Easy
19	List mein minimum element ka index dhundo.	*min track + index*	Easy
20	Unsorted list mein kisi range [a,b] ke beech ke saare elements print karo.	*if a<=x<=b*	Medium
21	2D matrix mein ek value dhundo aur uski (row, col) print karo.	*nested loop*	Medium
22	List mein pehla aur aakhri occurrence ka index dhundo.	*first/last flag*	Medium

⚡ ALGORITHM 2 — Binary Search

Key Concepts

- Binary Search = sorted array mein mid check karo, half eliminate karo. Repeat.
- Time Complexity: O(log n) — har step mein array half ho jaata hai.
- Space Complexity: O(1) iterative, O(log n) recursive.
- ZARURI: Array sorted hona chahiye, tab hi Binary Search kaam karta hai.
- 3 cases: arr[mid]==target (found), arr[mid]<target (right half), arr[mid]>target (left half).
- mid = (low + high) // 2 — overflow se bachne ke liye low + (high-low)//2 bhi use kar sakte ho.

Practice Questions

#	Question	Hint	Level
23	Sorted list mein ek element dhundo — index return karo, -1 agar na mile.	*low,high,mid loop*	Easy
24	Sorted list mein ek element ka pehla occurrence dhundo (Binary Search se).	*found pe high=mid-1*	Medium
25	Sorted list mein ek element ka aakhri occurrence dhundo.	*found pe low=mid+1*	Medium
26	Sorted list mein ek number insert karne ki sahi position dhundo.	*lower bound concept*	Medium
27	Sorted list mein peak element dhundo (neighbours se bada).	*arr[mid]>arr[mid+1]*	Medium
28	Rotated sorted array mein ek element dhundo. (e.g. [4,5,6,1,2,3])	*find pivot first*	Hard
29	Answer Binary Search: 1 se N tak ka square root nikalo bina sqrt() use kiye.	*binary search on answer*	Hard

□ ALGORITHM 3 — Two Pointer Technique

Key Concepts

- Two Pointer = do pointers (left, right) simultaneously chalao — ek start se, ek end se.
- Time Complexity: O(n) — single pass mein solve ho jaata hai.
- Space Complexity: O(1) — koi extra array nahi.
- Kab use karein: Sorted array mein pair/triplet dhundhna, palindrome check, reverse karna.
- Pattern: while left < right — condition dekho — left++ ya right-- karo.

Practice Questions

#	Question	Hint	Level
30	Sorted list mein do numbers dhundo jinki sum = target. (Two Pointer se)	*left++, right--*	Easy
31	List ko in-place reverse karo Two Pointer se.	*swap arr[l] arr[r]*	Easy
32	String palindrome check karo Two Pointer se.	*l=0, r=n-1*	Easy
33	Sorted list mein duplicate elements remove karo in-place.	*slow-fast pointer*	Medium
34	List mein 0s, 1s aur 2s ko sort karo bina sort() use kiye. (Dutch National Flag)	*3 pointer: lo,mid,hi*	Medium
35	Sorted list mein triplet dhundo jinki sum = 0.	*fix one + two pointer*	Medium
36	Do sorted lists ko merge karo O(1) extra space mein.	*two pointer on both*	Hard
37	Container with most water: list of heights hai, max water nikalo do heights ke beech.	*max(min(h[l],h[r])*w)*	Hard

ALGORITHM 4 — Sliding Window

Key Concepts

- Sliding Window = ek fixed ya variable size window array pe slide karo.
- Time Complexity: O(n) — O(n^2) brute force ko O(n) mein convert karta hai.
- Fixed Window: size fixed hoti hai — window slide karo, start+1, end+1.
- Variable Window: condition ke hisaab se window expand ya shrink hoti hai.
- Kab use karein: Subarray ka max/min sum, longest subarray with condition.

Practice Questions

#	Question	Hint	Level
38	Size k ka subarray dhundo jiska sum maximum ho.	*fixed window*	Easy
39	Size k ka subarray dhundo jiska average maximum ho.	*sum/k*	Easy
40	Longest subarray dhundo jisme saare elements equal hon.	*variable window*	Medium
41	Longest subarray dhundo jisme sum <= k ho.	*shrink from left*	Medium
42	String mein longest substring dhundo bina repeating characters ke.	*set + left pointer*	Medium
43	String mein ek given pattern ke saare anagrams ke starting indexes nikalo.	*freq map compare*	Hard
44	Minimum size subarray dhundo jiska sum >= target ho.	*shrink window greedily*	Hard

➕ ALGORITHM 5 — Prefix Sum

Key Concepts

- Prefix Sum = pehle ek prefix array banao jisme prefix[i] = arr[0]+arr[1]+...+arr[i].
- Range Sum Query: sum(l,r) = prefix[r] - prefix[l-1]. O(1) mein!
- Preprocessing: O(n) time, phir har query O(1).
- Kab use karein: Jab baar baar subarray sum poochha jaaye.
- 2D Prefix Sum bhi hota hai — matrix queries ke liye.

Practice Questions

#	Question	Hint	Level
45	Prefix sum array banao given array se.	*prefix[i]=prefix[i-1]+arr[i]*	Easy
46	Prefix sum use karke range [l,r] ka sum O(1) mein nikalo.	*prefix[r]-prefix[l-1]*	Easy
47	Subarray dhundo jiska sum = k. (Prefix sum + HashMap)	*sum-k in map*	Medium

48	Longest subarray dhundo jiska sum = 0.	*prefix sum + first index map*	Medium
49	Array mein equal 0s aur 1s wala longest subarray dhundo.	*0 ko -1 banao + prefix*	Hard

□ ALGORITHM 6 — Sorting Algorithms

Key Concepts

- Bubble Sort: adjacent elements compare aur swap karo. O(n²) time, O(1) space. Stable.
- Selection Sort: har pass mein minimum dhundo aur sahi jagah rakho. O(n²) time. Not stable.
- Insertion Sort: har element ko sahi position pe insert karo. O(n²) worst, O(n) best. Stable.
- Merge Sort: Divide & Conquer — half karo, sort karo, merge karo. O(n log n) time, O(n) space. Stable.
- Quick Sort: Pivot choose karo, partition karo. O(n log n) avg, O(n²) worst. Not stable.
- Kab kya use karein: Small data → Insertion Sort. Large data → Merge/Quick Sort.

Practice Questions

#	Question	Hint	Level
50	Bubble Sort implement karo.	*adjacent swap*	Easy
51	Selection Sort implement karo.	*min find + swap*	Easy
52	Insertion Sort implement karo.	*shift and insert*	Easy
53	Merge Sort implement karo (recursion use karo).	*divide+merge*	Medium
54	Quick Sort implement karo.	*pivot+partition*	Medium
55	Counting Sort implement karo (sirf positive integers ke liye).	*freq array*	Medium
56	Kth smallest element dhundo array mein bina full sort kiye.	*partial sort / quickselect*	Hard
57	Nearly sorted array sort karo efficiently (har element max k positions se dur hai).	*insertion sort O(nk)*	Hard

□ MIXED — Interview Level Problems

Key Concepts

- Yeh problems multiple concepts combine karti hain — jaise Two Pointer + Sorting, ya Prefix Sum + HashMap.
- Pehle brute force socho, phir optimize karo. Time complexity batana mat bhoolo.
- Yeh problems almost har placement exam mein aate hain — inhe zaroor solve karo.

Practice Questions

#	Question	Hint	Level
58	Array mein majority element dhundo (n/2 se zyada baar aane wala). — Boyer-Moore Voting	*count+cancel*	**Medium**
59	Stock buy-sell: ek baar buy/sell karo — max profit nikalo.	*min so far track*	**Easy**
60	Stock buy-sell: multiple transactions allowed — max profit nikalo.	*every uphill add karo*	**Medium**
61	Subarray with maximum sum nikalo. — Kadane's Algorithm	*max_so_far, max_here*	**Medium**
62	Array ko k positions left rotate karo in-place.	*reverse 3 times*	**Medium**
63	Array mein missing number dhundo (1 se n tak, ek missing hai).	*n*(n+1)/2 - sum*	**Easy**
64	Array mein ek number twice aata hai baaki sab ek baar — duplicate dhundo.	*XOR all elements*	**Medium**
65	Merge 2 sorted arrays in-place without extra space.	*two pointer from end*	**Hard**
66	Next permutation nikalo array ki.	*find dip + swap + reverse*	**Hard**
67	Spiral order mein matrix print karo.	*4 boundaries shrink*	**Hard**
68	Longest increasing subsequence ki length nikalo.	*dp[i]=max(dp[j]+1)*	**Hard**
69	Array mein do elements ka maximum product nikalo (negative numbers bhi hain).	*max2 + min2*	**Medium**
70	Rearrange array: positive aur negative alternate karo (extra space allowed).	*2 lists + merge*	**Medium**

All the best for your placements! Keep solving, keep growing. □